

The Source-Term Surrogate for Stiff Chemical Kinetics

Why the flow map is learnable where the vector field is not

Abstract

Physics-informed neural networks fail on stiff chemical kinetics for a reason that is structural rather than incidental: the training residual inherits the Lipschitz constant of the chemical source term, which for hydrocarbon and hydrogen mechanisms spans eight to twelve decades. This note derives that obstruction, then shows that the object a working surrogate actually learns — the finite-time *flow map* of the chemistry substep — is a contraction precisely in the directions that make the vector field stiff. Stiffness, which destroys the residual formulation, is what makes the flow-map formulation well-posed. We then give the constraints that make such a surrogate usable inside a CFD solver: exact element and mass conservation by linear projection, temperature recovered from enthalpy rather than predicted, positivity, and per-species scale separation. We close with an evaluation protocol designed against the documented failure mode of this literature.

1 Where the surrogate goes

Consider a reacting flow with n_s species. Writing ρ for density, $\mathbf{u}$ for velocity and Y_k for the mass fraction of species k ,

$$\frac{\partial(\rho Y_k)}{\partial t} + \nabla \cdot (\rho \mathbf{u} Y_k) = \nabla \cdot (\rho D_k \nabla Y_k) + \dot{\omega}_k, \quad k = 1, \dots, n_s, \quad (1)$$

where $\dot{\omega}_k$ is the chemical production rate. Essentially every production solver advances (1) by operator splitting. In Strang form, over a step Δt ,

$$\Phi_{\Delta t} = \mathcal{T}_{\Delta t/2} \circ \mathcal{C}_{\Delta t} \circ \mathcal{T}_{\Delta t/2}, \quad (2)$$

with $\mathcal{T}$ the transport (advection–diffusion) operator and $\mathcal{C}$ the chemistry operator. The essential fact is that $\mathcal{C}_{\Delta t}$ is *spatially local*: in each cell it is an initial-value problem in the thermochemical state alone,

$$\phi = (T, Y_1, \dots, Y_{n_s})^\top \in \mathbb{R}^{n_s+1}, \quad \frac{d\phi}{dt} = \mathbf{S}(\phi), \quad (3)$$

with no spatial coupling and no neighbours. The chemistry substep is therefore the same $\mathbb{R}^{n_s+1} \rightarrow \mathbb{R}^{n_s+1}$ problem solved independently in every cell, 10^6 to 10^9 times per simulation. In a detailed-chemistry LES it routinely consumes 70–90% of wall time.

Definition 1.1 (Reaction map). The *reaction map* at step Δt is the time- Δt flow of (3),

$$\mathcal{R}_{\Delta t} : \mathbb{R}^{n_s+1} \rightarrow \mathbb{R}^{n_s+1}, \quad \mathcal{R}_{\Delta t}(\phi_0) = \phi_0 + \int_0^{\Delta t} \mathbf{S}(\phi(\tau)) d\tau, \quad (4)$$

where $\phi(\cdot)$ solves (3) from $\phi(0) = \phi_0$.

What is being replaced

The surrogate replaces *one call* to the stiff ODE integrator — CVODE, Radau, or a specialised implicit method — with one forward pass. It does not replace the CFD solver, the transport operator, the pressure solve, or the turbulence closure. Those remain exactly as they were, and they continue to enforce the conservation laws they always enforced. This is the entire reason the approach is tractable, and it is what distinguishes it from a physics-informed network that attempts to represent the solution field itself.

2 Why the residual formulation fails

2.1 The stiffness of the vector field

Let $J(\phi) = \partial \mathbf{S} / \partial \phi \in \mathbb{R}^{(n_s+1) \times (n_s+1)}$ be the Jacobian of the chemical source, with eigenvalues $\{\lambda_i\}$. Define the stiffness ratio on a trajectory,

$$\varsigma = \frac{\max_i |\operatorname{Re} \lambda_i|}{\min_{i: \operatorname{Re} \lambda_i \neq 0} |\operatorname{Re} \lambda_i|}. \quad (5)$$

For a hydrogen–air mechanism $\varsigma \sim 10^8$; for detailed hydrocarbon mechanisms ς reaches 10^{12} . The fast eigenvalues belong to radical species — H, O, OH, HO₂ — whose concentrations equilibrate on timescales of 10^{-10} to 10^{-8} s, while the slow modes governing heat release and fuel consumption evolve on 10^{-4} to 10^{-2} s.

2.2 The obstruction

A physics-informed network posits $\phi_\theta(t)$ represented by a network with parameters θ , and minimises the residual

$$L(\theta) = \int_0^T \left\| \frac{d\phi_\theta}{dt}(t) - \mathbf{S}(\phi_\theta(t)) \right\|^2 dt. \quad (6)$$

Proposition 2.1 (Residual conditioning). *Let $g(\theta) = \frac{d\phi_\theta}{dt} - \mathbf{S}(\phi_\theta)$ denote the residual and let $P_\theta = \partial \phi_\theta / \partial \theta$ be the parameter Jacobian. The Gauss–Newton approximation to the Hessian of (6) is*

$$H \approx \int_0^T P_\theta^\top (\partial_t - J)^\top (\partial_t - J) P_\theta dt. \quad (7)$$

The spectrum of $(\partial_t - J)^\top (\partial_t - J)$ contains the squared eigenvalue magnitudes $|\lambda_i|^2$. Hence, absent preconditioning,

$$\kappa(H) \gtrsim \varsigma^2, \quad (8)$$

and first-order optimisation converges at a rate governed by $1 - \mathcal{O}(\kappa(H)^{-1})$.

For $\varsigma = 10^8$ this gives $\kappa \gtrsim 10^{16}$, which is the reciprocal of double-precision epsilon. No learning-rate schedule, adaptive optimiser, or loss reweighting repairs a condition number at that magnitude; the fast modes saturate the residual and the slow modes — the ones carrying the heat release one actually wants — contribute gradient components below numerical noise.

The same obstruction, in classical language

This is not a machine-learning pathology. It is the identical obstruction that forbids explicit time integration of stiff systems. An explicit method requires $\Delta t \lesssim 2/\max_i |\lambda_i|$, which for $\varsigma = 10^8$ means resolving 10^{-10} s steps to follow 10^{-2} s dynamics. The classical response was not a better explicit method: it was to change formulation, to implicit schemes that solve a linear system with J rather than stepping along it. Gradient descent on (6) is an explicit method in disguise, and it fails for the same reason.

3 Why the flow map is learnable

The surrogate does not learn $\mathbf{S}$. It learns $\mathcal{R}_{\Delta t}$. This is a different object with different conditioning, and the difference is the whole argument.

Proposition 3.1 (The flow map contracts the stiff directions). *Let $\mathbf{S}$ be linearised about an equilibrium ϕ^* on the slow manifold, $\mathbf{S}(\phi) \approx J(\phi - \phi^*)$ with J diagonalisable, $J = V\Lambda V^{-1}$, $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_{n_s+1})$ and $\text{Re } \lambda_i \leq 0$. Then*

$$\mathcal{R}_{\Delta t}(\phi) - \phi^* = V \exp(\Lambda \Delta t) V^{-1} (\phi - \phi^*), \quad (9)$$

so the singular values of $D\mathcal{R}_{\Delta t}$ in the eigenbasis are $e^{\text{Re } \lambda_i \Delta t}$. Consequently

$$\text{Lip}(\mathcal{R}_{\Delta t}) \leq \kappa(V) e^{\Delta t \max_i \text{Re } \lambda_i} \leq \kappa(V), \quad (10)$$

whereas $\text{Lip}(\mathbf{S}) = \|J\| \sim \max_i |\lambda_i|$.

Proof. Immediate from $\mathcal{R}_{\Delta t} = \exp(J\Delta t)$ in the linearised setting and the spectral mapping theorem. The bound on the Lipschitz constant follows since $\text{Re } \lambda_i \leq 0$ gives $e^{\text{Re } \lambda_i \Delta t} \leq 1$. $\square$

The content of Proposition 3.1 is worth stating plainly. In the fast directions, $|\lambda_i| \Delta t \gg 1$, so $e^{\lambda_i \Delta t} \approx 0$: the flow map *annihilates* exactly those directions in which the vector field is violent. What survives after Δt is motion on the slow manifold, whose dimension is typically 5 to 15 regardless of n_s . The map is smooth, bounded, non-expansive, and effectively low-dimensional.

The inversion

Stiffness is the obstruction for the residual formulation and the *gift* for the flow-map formulation. A large ς means the fast transients have already decayed within one Δt , so the function being approximated is a projection onto a low-dimensional attracting manifold. The stiffer the chemistry, the more the flow map contracts, and the easier it is to learn. This is why the same problem that defeats a PINN yields to a supervised surrogate, and it is not a matter of engineering effort or network capacity.

Remark 3.2. This is also why the idea long predates machine learning. In-situ adaptive tabulation and flamelet-manifold methods are, in this language, nearest-neighbour and low-rank approximations of the same $\mathcal{R}_{\Delta t}$. A neural surrogate is a better interpolant for the same object, not a new physical idea.

4 Constructing the surrogate

4.1 Target and scale separation

Predict the *increment*, not the state:

$$\Delta\phi = \mathcal{R}_{\Delta t}(\phi_0) - \phi_0. \quad (11)$$

Mass fractions span more than twenty decades, from $Y \sim 1$ for the bath gas to $Y \sim 10^{-20}$ for trace radicals, and a uniformly-weighted ℓ^2 loss on raw Y is dominated entirely by N_2 . Two transformations are used, and they matter more than architecture.

Logarithmic (or Box–Cox) transform. With a floor $\epsilon > 0$,

$$\psi_k = \ln(Y_k + \epsilon), \quad \text{or} \quad \psi_k = \frac{(Y_k + \epsilon)^\nu - 1}{\nu}, \quad (12)$$

the latter reducing to the former as $\nu \rightarrow 0$. This makes relative error the quantity being minimised, which is the quantity that matters for a radical species four decades below the fuel.

Per-species standardisation. After transformation,

$$\tilde{\psi}_k = \frac{\psi_k - \mu_k}{\sigma_k}, \quad \mu_k = \mathbb{E}[\psi_k], \quad \sigma_k^2 = \text{Var}[\psi_k], \quad (13)$$

with the statistics computed over the training set. Each species then contributes comparably to the loss.

4.2 Exact conservation by projection

Let $E \in \mathbb{R}^{n_e \times n_s}$ have entries

$$E_{jk} = a_{jk} \frac{W_j}{W_k}, \quad (14)$$

where a_{jk} is the number of atoms of element j in species k , and W denotes molar mass. Element conservation during the chemistry substep is the *linear* statement

$$E \Delta \mathbf{Y} = \mathbf{0}, \quad (15)$$

and total mass conservation $\sum_k \Delta Y_k = 0$ is the additional row $\mathbf{1}^\top \Delta \mathbf{Y} = 0$. Append it to E . Rather than penalising (15) in the loss, impose it exactly: let $\mathcal{N}_\theta$ be the raw network output and project,

$$\boxed{\Delta \mathbf{Y} = (I - E^+ E) \mathcal{N}_\theta(\phi_0), \quad E^+ = E^\top (E E^\top)^{-1}.} \quad (16)$$

Since $(I - E^+ E)$ is the orthogonal projector onto $\ker E$, the constraint holds to machine precision for every input, trained or not. The cost is one $n_e \times n_s$ matrix product with $n_e \leq 5$, negligible against the forward pass, and E^+ is precomputed once.

Hard constraints beat penalties here

A conservation penalty $\lambda \|E \Delta \mathbf{Y}\|^2$ trades conservation against accuracy through a weight nobody can choose principledly, and it is violated on exactly the out-of-distribution states

where violation is most damaging. Projection (16) costs less, cannot be tuned wrong, and holds under distribution shift. Where a constraint is linear, project; reserve penalties for constraints that are not.

4.3 Temperature from enthalpy, not from the network

At constant pressure the chemistry substep conserves specific enthalpy:

$$h = \sum_{k=1}^{n_s} Y_k h_k(T) = \text{const}, \quad h_k(T) = h_k^\circ + \int_{T^\circ}^T c_{p,k}(T') dT'. \quad (17)$$

Do not give the network a temperature output. Predict $\Delta \mathbf{Y}$ only, then recover T by solving the scalar equation (17) for T with Newton’s method,

$$T^{(m+1)} = T^{(m)} - \frac{\sum_k Y_k h_k(T^{(m)}) - h}{\sum_k Y_k c_{p,k}(T^{(m)})}, \quad (18)$$

which converges in two or three iterations from the previous temperature. This reduces the output dimension by one, removes a large-magnitude output that would otherwise dominate the loss, and makes energy conservation exact rather than approximate.

4.4 Positivity and the simplex

Mass fractions must satisfy $Y_k \geq 0$ and $\sum_k Y_k = 1$. Predicting in the transformed variable (12) and inverting gives $Y_k > 0$ automatically. Renormalise afterwards,

$$Y_k \leftarrow \frac{\max(Y_k, 0)}{\sum_j \max(Y_j, 0)}, \quad (19)$$

applied *after* the projection (16) and accepted as a final $\mathcal{O}(\text{machine } \epsilon)$ correction.

5 The failure mode that matters: a posteriori drift

A surrogate with excellent single-step error can still destroy a simulation. Writing $\hat{\mathcal{R}}$ for the surrogate and ε_n for the error after n applications, one step of error propagation gives

$$\varepsilon_{n+1} \leq \text{Lip}(\mathcal{R}_{\Delta t}) \varepsilon_n + \left\| \hat{\mathcal{R}} - \mathcal{R}_{\Delta t} \right\|_\infty. \quad (20)$$

By Proposition 3.1, $\text{Lip}(\mathcal{R}_{\Delta t}) \leq 1$ on the stiff directions, so error does *not* amplify along the attracting manifold — which is the structural reason the approach is stable at all. The danger is elsewhere: the coupled system revisits states the training set never covered, because transport carries cells into composition space that one-dimensional flame data does not populate. Two mitigations are standard.

1. **Manifold-aware sampling.** Draw training states from canonical flames — freely propagating premixed, counterflow diffusion, homogeneous ignition — which populate the attracting manifold densely.
2. **Physics-aware augmentation.** Interpolate between neighbouring sampled states and add constrained random perturbations that respect (15) and (17), thickening the training distribution transverse to the manifold without leaving the physically admissible set.

6 Evaluation protocol

The published record in this area carries a documented reporting problem: of articles claiming that a machine-learned method outperforms a standard numerical method on a fluid-related PDE, 79% (60 of 76) compared against a weak baseline, with outcome-reporting and publication bias both present.¹ The protocol below is written against that finding.

1. **The baseline is a tuned CVODE at the tolerance actually required**, not at a tolerance chosen to make it slow. Report the tolerance.
2. **Include the real incumbent.** The production alternative is not naive direct integration; it is tabulation — ISAT or a flamelet manifold. A surrogate that beats CVODE but loses to ISAT has not earned its complexity.
3. **Report a posteriori, not a priori.** Single-step L^2 error on a held-out set is not the claim. The claim is a coupled CFD run: ignition delay, laminar flame speed, extinction strain rate, and the full temperature and major-species profiles against the direct-integration reference.
4. **Report end-to-end wall time**, including the projection, the Newton solve for T , and the host-device transfers — not kernel time.
5. **State the conservation residuals** $\|E\Delta Y\|_\infty$ and the enthalpy drift over the whole run.
6. **Report where it breaks.** Name the equivalence ratio, pressure and temperature range outside which the surrogate departs from the reference, and publish that boundary with the model.

The honest target

Reported gains for this formulation are $\mathcal{O}(10^2)$ on isolated chemistry evaluation against CVODE and roughly 10–20 $\times$ end-to-end in coupled CFD. The gap between those two numbers is the point: once chemistry stops dominating, Amdahl’s law caps the benefit, and further speedup must come from transport. A surrogate is worth building when a design sweep or an uncertainty quantification study needs 10^5 – 10^6 runs and tolerates percent-level error. It is not worth building to make a single high-fidelity run faster.

¹N. McGreivy and A. Hakim, *Weak baselines and reporting biases lead to overoptimism in machine learning for fluid-related partial differential equations*, Nature Machine Intelligence **6**, 1256–1269 (2024).